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Inventor(s)	Locke; Christopher Brian et al.

Hybrid silicone and acrylic adhesive cover for use with wound treatment

Abstract

Sealing members and methods of manufacturing the same are described. A first film layer and a second film layer each having a first side and a second side can be provided. A first adhesive can be coupled to the second side of the first film layer to form a first adhesive layer. A second adhesive can be coupled to the second side of the second film layer to form a second adhesive layer, and a third adhesive can be coupled to the first side of the second film layer to form a third adhesive layer. One or more perforations can be formed through the third adhesive layer, the second film layer, and the second adhesive layer. A first side of the third adhesive layer can be coupled to a second side of the first adhesive layer.

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Background/Summary

RELATED APPLICATIONS (1) This application is a Divisional of U.S. patent application Ser. No. 15/265,718, entitled “Hybrid Silicone and Acrylic Adhesive Cover for use with Wound Treatment,” filed Sep. 14, 2016, which claims the benefit under 35 USC 119(e), of the filing of U.S. Provisional Patent Application No. 62/220,064, entitled “Hybrid Silicone and Acrylic Adhesive Cover for use

with Wound Treatment,” filed Sep. 17, 2015, which are incorporated herein by reference for all purposes.

TECHNICAL FIELD

(1) The invention set forth in the appended claims relates generally to tissue treatment systems and more particularly, but without limitation, to a cover for use with treatment of a tissue site.

BACKGROUND

(2) Clinical studies and practice have shown that reducing pressure in proximity to a tissue site can augment and accelerate growth of new tissue at the tissue site. The applications of this phenomenon are numerous, but it has proven particularly advantageous for treating wounds. Regardless of the etiology of a wound, whether trauma, surgery, or another cause, proper care of the wound is important to the outcome. Treatment of wounds or other tissue with reduced pressure may be commonly referred to as “negative-pressure therapy,” but is also known by other names, including “negative-pressure wound therapy,” “reduced-pressure therapy,” “vacuum therapy,” “vacuum-assisted closure,” and “topical negative-pressure,” for example. Negative-pressure therapy may provide a number of benefits, including migration of epithelial and subcutaneous tissues, improved blood flow, and micro-deformation of tissue at a wound site. Together, these benefits can increase development of granulation tissue and reduce healing times.

(3) While the clinical benefits of negative-pressure therapy are widely known, the cost and complexity of therapy can be a limiting factor in its application, and the development and operation of therapy systems, components, and processes continues to present significant benefits to healthcare providers and patients.

SUMMARY

(4) New and useful systems, apparatuses, and methods for providing a sealed environment for negative-pressure therapy are set forth in the appended claims. Illustrative embodiments are also provided to enable a person skilled in the art to make and use the claimed subject matter.

(5) For example, in some embodiments, a method of manufacturing a sealing member is described. A first film layer and a second film layer each having a first side and a second side can be provided. A first adhesive can be coupled to the second side of the first film layer to form a first adhesive layer. A first side of the first adhesive layer can be coupled to the second side of the first film layer. A second adhesive can be coupled to the second side of the second film layer to form a second adhesive layer. A first side of the second adhesive layer can be coupled to the second side of the second film layer. One or more perforations can be formed through the second film layer and the second adhesive layer. The first side of the second film layer can be coupled to a second side of the first adhesive layer.

(6) More generally, a cover for a dressing of a negative-pressure therapy system is described. A bonding adhesive can be coupled to a first side of a first elastomeric film to form a bonding adhesive layer having a first side adjacent to the first side of the first elastomeric film. A sealing adhesive can be coupled to a first side of a second elastomeric film to form a sealing adhesive layer having a first side adjacent to the first side of the second elastomeric film. One or more apertures may be formed through the second elastomeric film and the sealing adhesive layer; and a second side of the sealing adhesive layer can be coupled to a second side of the bonding adhesive layer.

(7) Alternatively, other example embodiments may describe a sealing member having a first elastomeric film having a first side and a second side, a bonding adhesive layer coupled to the first elastomeric film, a second elastomeric film having a first side and a second side and coupled to the bonding adhesive layer, and a sealing adhesive layer coupled to the second elastomeric film. The sealing adhesive layer and the second elastomeric film may have a plurality of perforations extending through the second elastomeric film and the sealing adhesive layer. The sealing member may be formed by coupling a bonding adhesive to the second side of the first elastomeric film to form the bonding adhesive layer having a first side coupled to the second side of the first

elastomeric film and a second side. A sealing adhesive may be coupled to the second side of the second elastomeric film to form the sealing adhesive layer having a first side coupled to the second side of the second elastomeric film and a second side. One or more perforations may be formed through the second elastomeric film and the sealing adhesive layer. The second side of the bonding adhesive layer may be positioned proximate to the first side of the second elastomeric film, and the first side of the second elastomeric film may be coupled to the second side of the bonding adhesive layer.

(8) A tissue cover is also described herein. The tissue cover can include a first film layer having a first side and a second side, a first adhesive layer coupled to the first film layer, and a second adhesive layer coupled to the first adhesive, the second adhesive layer having a plurality of perforations extending through the second adhesive layer. The tissue cover can be formed by coupling a first adhesive to the second side of the first film layer to form the first adhesive layer having a first side adjacent to the second side of the first film layer. A second adhesive can be coupled to a second film layer to form the second adhesive layer having a first side adjacent to the second film layer. One or more perforations can be formed through the second film layer and the second adhesive layer; and a second side of the second adhesive layer can be coupled to a second side of the first adhesive layer.

(9) Objectives, advantages, and a preferred mode of making and using the claimed subject matter may be understood best by reference to the accompanying drawings in conjunction with the following detailed description of illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a sectional view with a portion shown in elevation of an example embodiment of a therapy system that can provide a sealed therapeutic environment in accordance with this specification;

(2) FIG. 1A is a detail view of a portion of a cover of the therapy system of FIG. 1 illustrating additional details that may be associated with some embodiments;

(3) FIG. 2 is a perspective exploded view of the cover of the therapy system of FIG. 1;

(4) FIG. 3 is a perspective partially assembled view of the cover of FIG. 2 illustrating additional details that may be associated with some embodiments;

(5) FIG. 4A is a sectional view of the cover of FIG. 2 illustrating additional details that may be associated with the cover in a first position;

(6) FIG. 4B is a sectional view of the cover of FIG. 2 illustrating additional details that may be associated with the cover in a second position;

(7) FIGS. 5A-5F are schematic views illustrating additional details that may be associated with an assembly process of the cover of FIG. 2;

(8) FIG. 6 is a perspective view illustrating additional details that may be associated with another cover that may be used with the therapy system of FIG. 1;

(9) FIG. 7 is a sectional view illustrating additional details that may be associated with the cover of FIG. 6;

(10) FIGS. 8A-8G are schematic views illustrating additional details that may be associated with an assembly process of the cover of FIG. 6;

(11) FIG. 9A is a perspective view illustrating additional details that may be associated with another cover that may be used with the therapy system of FIG. 1;

(12) FIG. 9B is a sectional view illustrating additional details that may be associated with the cover of FIG. 9A;

(13) FIG. 9C is a sectional view illustrating additional details that may be associated with the cover

of FIG. 9C;

(14) FIG. 10 is a perspective exploded view of another cover that can be used with the therapy system of FIG. 1;

(15) FIG. 11 is a perspective exploded view of another cover that can be used with the therapy system of FIG. 1;

(16) FIG. 12A is a sectional view of the cover of FIG. 11 illustrating additional details that may be associated with the cover in a first position;

(17) FIG. 12B is a sectional view of the cover of FIG. 11 illustrating additional details that may be associated with the cover in a second position;

(18) FIGS. 13A-13G are schematic views illustrating additional details that may be associated with an assembly process of the cover of FIG. 11;

(19) FIG. 14 is a sectional view of the cover of FIG. 11 illustrating additional details associated with light transmission through the cover;

(20) FIG. 15 is a graph of visible light absorption for various covers, including the cover of FIG. 11;

(21) FIG. 16 is a graph of leak rate versus time for various covers, including the cover of FIG. 11;

(22) FIG. 17 is another graph of leak rate versus time for various covers, including the cover of FIG. 11; and

(23) FIG. 18 is a graph depicting maceration tendency versus volume of fluid instilled in a tissue site sealed with the cover of FIG. 11.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(24) The following description of example embodiments provides information that enables a person skilled in the art to make and use the subject matter set forth in the appended claims, but may omit certain details already well-known in the art. The following detailed description is, therefore, to be taken as illustrative and not limiting.

(25) The example embodiments may also be described herein with reference to spatial relationships between various elements or to the spatial orientation of various elements depicted in the attached drawings. In general, such relationships or orientation assume a frame of reference consistent with or relative to a patient in a position to receive treatment. However, as should be recognized by those skilled in the art, this frame of reference is merely a descriptive expedient rather than a strict prescription.

(26) The term “tissue site” in this context broadly refers to a wound, defect, or other treatment target located on or within tissue, including but not limited to, bone tissue, adipose tissue, muscle tissue, neural tissue, dermal tissue, vascular tissue, connective tissue, cartilage, tendons, or ligaments. A wound may include chronic, acute, traumatic, subacute, and dehiscent wounds, partial-thickness burns, ulcers (such as diabetic, pressure, or venous insufficiency ulcers), flaps, and grafts, for example. The term “tissue site” may also refer to areas of any tissue that are not necessarily wounded or defective, but are instead areas in which it may be desirable to add or promote the growth of additional tissue. For example, negative pressure may be applied to a tissue site to grow additional tissue that may be harvested and transplanted.

(27) FIG. 1 is a sectional view with a portion shown in elevation of an example embodiment of a therapy system 100 for negative-pressure therapy in accordance with this specification. The therapy system 100 may include a negative-pressure supply, such as a negative-pressure source 104, and may include or be configured to be coupled to a distribution component, such as a dressing 102. In general, a distribution component may refer to any complementary or ancillary component configured to be fluidly coupled to a negative-pressure supply in a fluid path between a negative-pressure supply and a tissue site. A distribution component is preferably detachable, and may be disposable, reusable, or recyclable. For example, the dressing 102 may be fluidly coupled to the negative-pressure source 104, as illustrated in FIG. 1. A dressing may include a tissue cover, a tissue interface, or both in some embodiments. The dressing 102, for example, may include a cover

106 and a tissue interface **108**. A regulator or a controller may also be coupled to the negative-pressure source **104**.

(28) In some embodiments, a dressing interface may facilitate coupling the negative-pressure source **104** to the dressing **102**. For example, a dressing interface may be a T.R.A.C.® Pad or Sensa T.R.A.C.® Pad available from Kinetic Concepts, Inc. of San Antonio, Texas.

(29) The therapy system **100** may optionally include a fluid container coupled to or integral with the dressing **102** and to the negative-pressure source **104**. A fluid container may include a container, canister, pouch, or other storage component, which can be used to manage exudates and other fluids withdrawn from a tissue site. In many environments, a rigid container may be preferred or required for collecting, storing, and disposing of fluids. In other environments, fluids may be properly disposed of without rigid container storage, and a re-usable container could reduce waste and costs associated with negative-pressure therapy.

(30) Additionally, the therapy system **100** may include sensors to measure operating parameters and provide feedback signals to a controller of the negative-pressure source **104**. The feedback signals can be indicative of the operating parameters, such as pressure at the tissue site, humidity at the tissue site, or temperature at the tissue site, for example. The therapy system **100** may include a pressure sensor, a humidity sensor, a temperature sensor, or the like. In some embodiments, the sensors may be electrical sensors that can communicate with the therapy system **100** through electric signals. In other embodiments, the sensors may be pneumatically or hydraulically operated. A pressure sensor may also be coupled or configured to be coupled to a distribution component and to the negative-pressure source **104**.

(31) Components may be fluidly coupled to each other to provide a path for transferring fluids (i.e., liquid and/or gas) between the components. For example, components may be fluidly coupled through a fluid conductor, such as a tube **107**. A “tube,” as used herein, broadly includes a tube, pipe, hose, conduit, or other structure with one or more lumina adapted to convey a fluid between two ends. Typically, a tube is an elongated, cylindrical structure with some flexibility, but the geometry and rigidity may vary. In some embodiments, components may also be coupled by virtue of physical proximity, being integral to a single structure, or being formed from the same piece of material. Moreover, some fluid conductors may be molded into or otherwise integrally combined with other components. Coupling may also include mechanical, thermal, electrical, or chemical coupling (such as a chemical bond) in some contexts. For example, a tube may mechanically and fluidly couple the dressing **102** to a container.

(32) In general, components of the therapy system **100** may be coupled to each other directly or indirectly. For example, the negative-pressure source **104** may be directly coupled to the dressing **102** or indirectly coupled to the dressing **102** through a container.

(33) The fluid mechanics of using a negative-pressure source to reduce pressure in another component or location, such as within a sealed therapeutic environment, can be mathematically complex. However, the basic principles of fluid mechanics applicable to negative-pressure therapy are generally well-known to those skilled in the art, and the process of reducing pressure may be described illustratively herein as “delivering,” “distributing,” or “generating” negative pressure, for example.

(34) In general, exudates and other fluids flow toward lower pressure along a fluid path. Thus, the term “downstream” typically implies a position in a fluid path relatively closer to a source of negative pressure or further away from a source of positive pressure. Conversely, the term “upstream” implies a position relatively further away from a source of negative pressure or closer to a source of positive pressure. Similarly, it may be convenient to describe certain features in terms of fluid “inlet” or “outlet” in such a frame of reference. This orientation is generally presumed for purposes of describing various features and components herein. However, the fluid path may also be reversed in some applications (such as by substituting a positive-pressure source for a negative-pressure source) and this descriptive convention should not be construed as a limiting convention.

(35) “Negative pressure” generally refers to a pressure less than a local ambient pressure, such as the ambient pressure in a local environment external to a sealed therapeutic environment provided by the dressing **102**. In many cases, the local ambient pressure may also be the atmospheric pressure at which a tissue site is located. Alternatively, the pressure may be less than a hydrostatic pressure associated with tissue at the tissue site. Unless otherwise indicated, values of pressure stated herein are gauge pressures. Similarly, references to increases in negative pressure typically refer to a decrease in absolute pressure, while decreases in negative pressure typically refer to an increase in absolute pressure. While the amount and nature of negative pressure applied to a tissue site may vary according to therapeutic requirements, the pressure is generally a low vacuum, also commonly referred to as a rough vacuum, between -5 mmHg (-667 Pa) and -500 mmHg (-66.7 kPa). Common therapeutic ranges are between -75 mmHg (-9.9 kPa) and -300 mmHg (-39.9 kPa).

(36) A negative-pressure supply, such as the negative-pressure source **104**, may be a reservoir of air at a negative pressure, or may be a manual or electrically-powered device that can reduce the pressure in a sealed volume, such as a vacuum pump, a suction pump, a wall suction port available at many healthcare facilities, or a micro-pump, for example. A negative-pressure supply may be housed within or used in conjunction with other components, such as sensors, processing units, alarm indicators, memory, databases, software, display devices, or user interfaces that further facilitate therapy. For example, in some embodiments, the negative-pressure source **104** may be combined with a controller and other components into a therapy unit. A negative-pressure supply may also have one or more supply ports configured to facilitate coupling and de-coupling the negative-pressure supply to one or more distribution components.

(37) The tissue interface **108** can be generally adapted to contact a tissue site. The tissue interface **108** may be partially or fully in contact with the tissue site. If the tissue site is a wound, for example, the tissue interface **108** may partially or completely fill the wound, or may be placed over the wound. The tissue interface **108** may take many forms, and may have many sizes, shapes, or thicknesses depending on a variety of factors, such as the type of treatment being implemented or the nature and size of a tissue site. For example, the size and shape of the tissue interface **108** may be adapted to the contours of deep and irregular shaped tissue sites. Moreover, any or all of the surfaces of the tissue interface **108** may have projections or an uneven, coarse, or jagged profile that can induce strains and stresses on a tissue site, which can promote granulation at the tissue site.

(38) In some embodiments, the tissue interface **108** may be a manifold. A “manifold” in this context generally includes any substance or structure providing a plurality of pathways adapted to collect or distribute fluid across a tissue site under pressure. For example, a manifold may be adapted to receive negative pressure from a negative-pressure supply and distribute negative pressure through multiple apertures across a tissue site, which may have the effect of collecting fluid from across a tissue site and drawing the fluid toward a negative-pressure supply. In some embodiments, the fluid path may be reversed or a secondary fluid path may be provided to facilitate delivering fluid across a tissue site.

(39) In some illustrative embodiments, the pathways of a manifold may be interconnected to improve distribution or collection of fluids across a tissue site. In some illustrative embodiments, a manifold may be a porous foam material having interconnected cells or pores. For example, cellular foam, open-cell foam, reticulated foam, porous tissue collections, and other porous material such as gauze or felted mat generally include pores, edges, and/or walls adapted to form interconnected fluid channels. Liquids, gels, and other foams may also include or be cured to include apertures and fluid pathways. In some embodiments, a manifold may additionally or alternatively comprise projections that form interconnected fluid pathways. For example, a manifold may be molded to provide surface projections that define interconnected fluid pathways.

(40) The average pore size of a foam may vary according to needs of a prescribed therapy. For example, in some embodiments, the tissue interface **108** may be a foam having pore sizes in a

range of about 400 to about 600 microns. The tensile strength of the tissue interface **108** may also vary according to the needs of a prescribed therapy. For example, the tensile strength of a foam may be increased for instillation of topical treatment solutions. In one non-limiting example, the tissue interface **108** may be an open-cell, reticulated polyurethane foam such as GranuFoam® dressing or VeraFlo® foam, both available from Kinetic Concepts, Inc. of San Antonio, Texas. (41) The tissue interface **108** may be either hydrophobic or hydrophilic. In an example in which the tissue interface **108** may be hydrophilic, the tissue interface **108** may also wick fluid away from a tissue site, while continuing to distribute negative pressure to the tissue site. The wicking properties of the tissue interface **108** may draw fluid away from a tissue site by capillary flow or other wicking mechanisms. An example of a hydrophilic foam is a polyvinyl alcohol, open-cell foam such as V.A.C. WhiteFoam® dressing available from Kinetic Concepts, Inc. of San Antonio, Texas. Other hydrophilic foams may include those made from polyether. Other foams that may exhibit hydrophilic characteristics include hydrophobic foams that have been treated or coated to provide hydrophilicity.

(42) The tissue interface **108** may further promote granulation at a tissue site when pressure within a sealed therapeutic environment is reduced. For example, any or all of the surfaces of the tissue interface **108** may have an uneven, coarse, or jagged profile that can induce microstrains and stresses at a tissue site if negative pressure is applied through the tissue interface **108**.

(43) In some embodiments, the tissue interface **108** may be constructed from bioresorbable materials. Suitable bioresorbable materials may include, without limitation, a polymeric blend of polylactic acid (PLA) and polyglycolic acid (PGA). The polymeric blend may also include without limitation polycarbonates, polyfumarates, and caprolactones. The tissue interface **108** may further serve as a scaffold for new cell-growth, or a scaffold material may be used in conjunction with the tissue interface **108** to promote cell-growth. A scaffold is generally a substance or structure used to enhance or promote the growth of cells or formation of tissue, such as a three-dimensional porous structure that provides a template for cell growth. Illustrative examples of scaffold materials include calcium phosphate, collagen, PLA/PGA, coral hydroxy apatites, carbonates, or processed allograft materials.

(44) In some embodiments, the cover **106** may be a sealing member configured to provide a bacterial barrier and protection from physical trauma. The cover **106** may also be constructed from a material that can reduce evaporative losses and provide a fluid seal between two components or two environments, such as between a therapeutic environment and a local external environment. FIG. 1A is a detail view of a portion of the therapy system **100** of FIG. 1, illustrating additional details that may be associated with the cover **106**. The cover **106** may be, for example, a film layer **114** or membrane that can provide a seal adequate to maintain a negative pressure at a tissue site for a given negative-pressure source. The cover **106** may have a high moisture-vapor transmission rate (MVTR) in some applications. For example, the MVTR may be at least about 300 g/m.sup.2 per twenty-four hours. In some embodiments, the cover **106** may be a polymer cover, such as a polyurethane film, that is permeable to water vapor but impermeable to liquid. Such covers typically have a thickness in the range of about 25 to about 50 microns. For permeable materials, the permeability generally should be low enough that a desired negative pressure may be maintained.

(45) The cover **106** may include an attachment device to attach the cover **106** to an attachment surface, such as undamaged epidermis, a gasket, or another cover. The attachment device may take many forms. For example, an attachment device may be a medically-acceptable, pressure-sensitive adhesive that extends about a periphery, a portion, or an entire sealing member. In some embodiments, for example, some or all of the cover **106** may be coated with an acrylic adhesive having a coating weight between about 25 and about 65 grams per square meter (g.s.m.). Thicker adhesives, or combinations of adhesives, may be applied in some embodiments to improve the seal and reduce leaks. Other example embodiments of an attachment device may include a double-sided

tape, paste, hydrocolloid, hydrogel, silicone gel, or organogel.

(46) In some embodiments, the attachment device may comprise a first adhesive layer **110** and a second adhesive layer **112**. The first adhesive layer **110** may be a continuous adhesive layer and may be secured to the film layer **114**. The second adhesive layer **112** may be secured to the first adhesive layer **110** and have a plurality of apertures **116** extending through the second adhesive layer **112**. In some embodiments, a support layer **111** may be disposed between the second adhesive layer **112** and the first adhesive layer **110**. In some embodiments, the support layer **111** may have a plurality of apertures **132**.

(47) In operation, the tissue interface **108** may be placed within, over, on, or otherwise proximate to a tissue site. The cover **106** may be placed over the tissue interface **108** and sealed to an attachment surface near the tissue site. For example, the cover **106** may be sealed to undamaged epidermis peripheral to a tissue site. Thus, the dressing **102** can provide a sealed therapeutic environment proximate to a tissue site that is substantially isolated from the external environment. The negative-pressure source **104** can reduce the pressure in the sealed therapeutic environment. Negative pressure applied across the tissue site through the tissue interface **108** in the sealed therapeutic environment can induce macrostrain and microstrain in the tissue site, as well as remove exudates and other fluids from the tissue site, which can be collected in a container.

(48) In some embodiments, a battery or other portable power supply may supply power to the negative-pressure source **104**. Use of battery power can significantly decrease the total power available to a negative-pressure supply. As a result, power drains that would be considered negligible in a device powered through an electrical outlet connection may significantly reduce the ability of the negative-pressure supply to provide therapy. A power drain refers to operation of the negative-pressure supply that requires use of electrical power, for example, operation of a pump to generate negative pressure. Power drains may be caused by small dressing leaks, for example. A small dressing leak can drain power from a battery of a negative-pressure supply by repeatedly triggering operation of the negative-pressure supply to maintain the necessary negative pressure at the tissue site. These power drains shorten the useful life of the negative-pressure supply before disposal of the negative-pressure supply, recharge of the battery, or battery replacement is required. Leak detection techniques may help to identify some leaks that may then be sealed by the user; however, small leaks will challenge the most sensitive leak detection systems and may often go undetected.

(49) Small leaks may occur between the cover and the epidermis surrounding a tissue site if the cover fails to completely seal to the epidermis. Covers can be a balance between the strength of the attachment device required to enable the cover to seal against leaks and the pain which may result if the cover is removed. Generally, adhesives used to form the attachment device can be loosely classified as “bonding” adhesives or “sealing” adhesives.

(50) A bonding adhesive is generally stronger than a sealing adhesive. A bonding adhesive may also provide a suitable seal, but the strength of a bonding adhesive can cause significantly more discomfort upon removal. In addition, removing a cover with a bonding adhesive may cause significant damage to patients having delicate or damaged skin.

(51) A sealing adhesive can generally be characterized by lower strength and lower viscosity than a bonding adhesive. A cover that has a sealing adhesive can fill gaps between the cover and the epidermis to limit leaks and can be easy to remove with low discomfort to the patient. Various sealing, gap-filling adhesives, such as silicone, hydrocolloids, and hydrogels, have been tried but each has drawbacks. For example, hydrogel adhesives are usually low tack and prone to swelling, creep, and mobility when used with fluid systems. In another example, silicone adhesives can fill gaps and seal, but are not breathable and may lose the necessary mechanical bonding strength as the silicone adhesives interact with moisture during use. To counter these problems, silicone adhesives often require additional materials to secure the silicone adhesive to the patient. For example, a low-leak cover may be formed from two adhesive layers: a thick sealing adhesive,

perhaps in the shape of a gasket or ring, and a thinner bonding adhesive layer used to keep the sealing adhesive in place. The thinner bonding adhesive may be applied as cover strips, or combined with the thicker sealing adhesive as an outer border. Low-leak covers constructed in this way can be more complex than a cover using a single adhesive, increasing the complexity of manipulation and operation.

(52) A hybrid cover having a thick sealing layer that is perforated and laminated over an adhesive-coated film can overcome some of these challenges. For example, a hybrid cover may include a film layer having a bonding adhesive applied directly to the film layer, and a sealing adhesive applied directly to the bonding adhesive. The bonding adhesive and the sealing adhesive can be laminated to the film layer by applying the bonding adhesive and the sealing adhesive to the film layer in a liquid form and the curing the bonding adhesive and the sealing adhesive in place on the film layer. Curing may involve toughening or hardening of polymer material by cross-linking of polymer chains within the polymer material. Curing can include the addition of heat, chemical additives and ultraviolet radiation to cause cross-linking within the polymer material. After lamination, the sealing adhesive can be perforated to expose the bonding adhesive. If the cover is applied to a patient, the bonding adhesive can be pushed through the perforations of the sealing adhesive to secure the sealing adhesive to the patient. This laminated configuration may provide the benefits of the sealing adhesive and the bonding adhesive over the entire cover area. For example, the laminated configuration may be conformable and of sufficient strength to seal small leaks, and can be mechanically affixed to an epidermis without secondary processes. The laminated configuration can also be removable with minimal trauma to the patient.

(53) However, construction of a laminated attachment device may require a complicated manufacturing process. For example, to form the sealing layer, a thick sealing adhesive may be applied to a scrim layer. A scrim layer may be a sheet of strong, coarse fabric or webbing. The sealing adhesive may be cast onto the scrim layer and cured around the scrim layer. Casting involves pouring a liquid material, for example an adhesive, into a mold having a hollowed cavity of the desired shape. The liquid material is allowed to solidify or cure into a solid or semi-solid body. The scrim layer may provide support for the sealing adhesive and can aid in the manipulation of the sealing adhesive during the manufacturing process. After casting the sealing adhesive onto the scrim layer, the sealing adhesive and scrim layer may be coupled to an adhesive coated film layer to form a cover.

(54) Moreover, fabric in the scrim layer can inhibit the light transmittance of the cover by increasing the absorbance of the cover. For example, the scrim layer may be a barrier in the cover that can absorb light. The scrim layer may also reflect light transmitting through the cover, or scatter light reflecting through the cover, causing the light to be attenuated by the cover. As the light is absorbed, reflected or scattered, the light cannot reflect off of a wound covered by the cover. As a consequence, the scrim layer may cause the cover to appear opaque and prevent a clinician from seeing through the cover to view the dressing or tissue site.

(55) The scrim layer may also add additional thickness to the sealing layer. If the scrim layer is exposed, the exposed portions may prevent the sealing layer from forming an effective seal between the cover and the epidermis. As a result, additional sealing adhesive may be needed to ensure that the scrim layer is not exposed. The coarse fabric or webbing of a scrim layer may also have large openings between threads of the fabric or webbing. If the sealing adhesive is cast onto the scrim layer, the sealing adhesive may need to fill the openings in the fabric. As a result, if a scrim layer is used, even more sealing adhesive may be needed to completely cover the scrim layer and provide a surface for adhesion of the sealing adhesive to a patient. The scrim layer, along with any additional sealing adhesive needed, may add stiffness to the cover, reducing the conformability of the cover and inhibiting the ability of the sealing adhesive to migrate and seal to tissue.

(56) As disclosed herein, these challenges and others can be overcome with a cover having a hybrid configuration with a sealing adhesive layer and a bonding adhesive layer that can be constructed

without the use of a scrim layer. Light transmittance can also be improved to provide accurate cover positioning and visualization of the tissue interface and periwound during treatment. In some embodiments, flexibility and conformability can be increased, and adhesives used more efficiently when manufacturing the cover. Increased efficiency of adhesive use may enable a thinner coating of adhesive, which can decrease the cost.

(57) FIG. 2 is a perspective exploded view of an example embodiment of the cover **106**. The cover **106** may include the film layer **114**, the first adhesive layer **110** adjacent to the film layer **114**, the support layer **111** adjacent to the first adhesive layer **110**, and the second adhesive layer **112** adjacent to the support layer **111**. The film layer **114** may be formed from a range of medically approved films ranging in thickness from about 15 microns (μm) to about 50 microns (μm). The film layer **114** may comprise a suitable material or materials, such as the following: hydrophilic polyurethane (PU), cellulotics, hydrophilic polyamides, polyvinyl alcohol, polyvinyl pyrrolidone, hydrophilic acrylics, hydrophilic silicone elastomers, and copolymers of these. In some embodiments, the film layer **114** may be formed from a breathable cast matt polyurethane film sold by Expopack Advanced Coatings of Wrexham, United Kingdom, under the name INSPIRE™ 2301 or INSPIRE™ 2327.

(58) The film layer **114** may have a high moisture-vapor-transfer-rate (MVTR). A high MVTR may allow vapor to egress the sealed therapeutic environment through the film layer **114** and inhibit liquids from exiting the sealed therapeutic environment through the film layer **114**. In some embodiments, the MVTR of the film layer **114** may be greater than or equal to about 300 grams per meter squared per twenty-four hours ($\text{g/m.sup.2/24 hours}$). In other embodiments, the MVTR of the film layer **114** may be greater than or equal to about 1000 $\text{g/m.sup.2/24 hours}$. The illustrative INSPIRE™ 2301 film may have an MVTR (tested using the inverted cup technique) of about 14400 $\text{g/m.sup.2/24 hours}$ and may be about 30 microns thick. In other embodiments, a film having a low MVTR or that allows no vapor transfer might be used. The film layer **114** can also function as a barrier to microorganisms.

(59) The film layer **114** may have a first side **118** and a second side **120**. The first adhesive layer **110** may also have a first side **122** and a second side **124**. The first side **122** of the first adhesive layer **110** may face the second side **120** of the film layer **114**. In some embodiments, the first adhesive layer **110** may be a medically-acceptable, pressure-sensitive adhesive, glue, bonding agent, or cement. For example, the first adhesive layer **110** may be formed from or include a bonding adhesive. The bonding adhesive may be a high bond strength acrylic adhesive, pattern rubber adhesive, high-tack silicone adhesive, or polyurethane, for example. In some embodiments, the bond strength of the bonding adhesive may have a peel adhesion or resistance to being peeled from a stainless steel material between about 6 N/25 mm to about 10 N/25 mm on stainless steel substrate at 23° C. at 50% relative humidity based on the American Society for Testing and Materials (“ASTM”) standard ASTM D3330. In some embodiments, the bonding adhesive of the first adhesive layer **110** comprises an acrylic adhesive with a coating weight of about 10 grams/m.sup.2 (gsm) to about 70 grams/m.sup.2 (gsm). The first adhesive layer **110** may have a thickness **156**, and in some embodiments, the thickness **156** of the first adhesive layer **110** may be about 30 microns to about 60 microns.

(60) The support layer **111** may have a first side **123** and a second side **125**. In some embodiments, the first side **123** of the support layer **111** may face the second side **124** of the first adhesive layer **110**. The support layer **111** may be an elastomeric film and may have a high moisture-vapor-transfer-rate (MVTR) similar to the film layer **114**. The support layer **111** may be formed from a range of medically-approved films ranging in thickness from about 15 microns (μm) to about 50 microns (μm). The support layer **111** may comprise a suitable material or materials, such as the following: hydrophilic polyurethane (PU), cellulotics, hydrophilic polyamides, polyvinyl alcohol, polyvinyl pyrrolidone, hydrophilic acrylics, hydrophilic silicone elastomers, and copolymers of these. In some embodiments, the support layer **111** may be formed from a breathable cast matt

polyurethane film sold by Expopack Advanced Coatings of Wrexham, United Kingdom under the name INSPIRE 2301 or INSPIRE 2327.

(61) In some embodiments, the MVTR of the support layer **111** may be greater than or equal to about 300 g/m.sup.2/24 hours. In other embodiments, the MVTR of the support layer **111** may be greater than or equal to about 1000 g/m.sup.2/24 hours. The illustrative INSPIRE™ 2301 film may have an MVTR (inverted cup technique) of about 14400 g/m.sup.2/24 hours and may be about 30 microns thick. In other embodiments, a support layer having a low MVTR or that allows no vapor transfer might be used. The support layer **111** can also function as a barrier to liquids and microorganisms.

(62) In some embodiments, the support layer **111** may have a plurality of apertures **132**. Each aperture **132** of the plurality of apertures **132** may extend through the support layer **111** from the first side **123** to the second side **125**. The plurality of apertures **132** may be numerous shapes, including without limitation, circles, squares, stars, ovals, polygons, slits complex curves, rectilinear shapes, or triangles. Each aperture **132** of the plurality of apertures **132** may have an effective diameter. An effective diameter of a non-circular area is the diameter of a circular area having the same area as the non-circular area. The average effective diameter is typically in the range of about 4 mm to about 50 mm. The plurality of apertures **132** may have a uniform pattern or may be randomly distributed on the support layer **111**. In some embodiments, the apertures **132** may be distributed so that centers of adjacent apertures **132** may be separated by between about 5 mm and about 100 mm. In other embodiments, the apertures **132** may be distributed so that the centers of adjacent apertures **132** may be separated by about 10 mm.

(63) Regardless of the shape of the plurality of apertures **132**, the plurality of apertures **132** in the support layer **111** may leave void spaces in the support layer **111**. The percentage of void space of the plurality of apertures **132** may be equal to the percentage of the volume or surface area of the void spaces created by the plurality of apertures **132** to the total volume or surface area of the support layer **111**. In some embodiments, the percentage of void space may be between about 40% and about 75%. In other embodiments, the percentage of void space may be about 55%. The organization of the plurality of apertures **132** can also impact the percentage of void space. The plurality of apertures **132** may be formed by punching, cutting, melting, or drilling the support layer **111**.

(64) The second adhesive layer **112** may have a first side **126** and a second side **128**. The first side **126** of the second adhesive layer **112** may face the second side **125** of the support layer **111** and have a thickness **154**. The second adhesive layer **112** may be formed from or include a sealing adhesive. The sealing adhesive may be an adhesive having a low to medium tackiness, for example, a silicone polymer, polyurethane, or an additional acrylic adhesive. Generally, tackiness may be understood as a bond strength of an adhesive after a very short contact time between the adhesive and a substrate, such as less than 60 seconds. In some embodiments, the bond strength of the sealing adhesive may have a peel adhesion or resistance to being peeled from a stainless steel material between about 0.5 N/25 mm to about 1.5 N/25 mm on stainless steel substrate at 23° C. at 50% relative humidity based on ASTM D3330. The sealing adhesive may achieve its bond strength above after a contact time with the substrate of less than 60 seconds. In some embodiments, the sealing adhesive may have a coating weight between about 150 gsm and about 250 gsm, be about 100 microns to about 400 microns thick, and have a tackiness that may be about 30% to about 50% of the tackiness of the bonding adhesive that may be used in the first adhesive layer **110**. In some embodiments a catalyst may be added to the sealing adhesive to provide additional therapeutic properties; for example, a platinum catalyst or a sulfur catalyst may be added to the sealing adhesive to aid in treatment.

(65) In some embodiments, the second adhesive layer **112** may have a plurality of apertures **116** that extend through the second adhesive layer **112** from the first side **126** to the second side **128**. The plurality of apertures **116** may be, for example, circles, squares, stars, ovals, polygons, slits,

complex curves, rectilinear shapes, triangles, or other shapes. Each aperture **116** of the plurality of apertures **116** may have an effective diameter typically in the range of about 4 mm to about 50 mm. The plurality of apertures **116** may have a uniform pattern or may be randomly distributed on the second adhesive layer **112**. In some embodiments, the apertures **116** may be distributed so that centers of adjacent apertures **116** may be separated by between about 5 mm and about 100 mm. In other embodiments, the apertures **116** may be distributed so that the centers of adjacent apertures **116** may be separated by about 10 mm.

(66) Regardless of the shape of the plurality of apertures **116**, the plurality of apertures **116** in the second adhesive layer **112** may leave void spaces in the second adhesive layer **112**. The percentage of void space of the plurality of apertures **116** may be equal to the percentage of the volume or surface area of the void spaces created by the plurality of apertures **116** to the total volume or surface area of the second adhesive layer **112**. In some embodiments, the percentage of void space may be between about 40% and about 75%. In other embodiments, the percentage of void space may be about 55%. The organization of the plurality of apertures **116** can also impact the percentage of void space. The plurality of apertures **116** may be formed by punching, cutting, melting, or drilling, for example.

(67) FIG. 3 is a perspective view of the cover **106** partially assembled, illustrating additional details that may be associated with some embodiments. As shown in FIG. 3, the film layer **114** may be coupled to the first adhesive layer **110**. For example, the second side **120** of the film layer **114** may be in contact with and coupled to the first side **122** of the first adhesive layer **110**. The film layer **114** and the first adhesive layer **110** may be coextensive. For example, the second side **120** of the film layer **114** and the first side **122** of the first adhesive layer **110** may be in contact across the entirety of their respective surface areas. The film layer **114** may be coupled to the first adhesive layer **110** by bonding, adhering, welding, or cross-linking, for example.

(68) Similarly, the support layer **111** may be coupled to the second adhesive layer **112**. For example, the second side **125** of the support layer **111** may be in contact with and coupled to the first side **126** of the second adhesive layer **112**. The support layer **111** and the second adhesive layer **112** may be coextensive. For example, the first side **126** of the second adhesive layer **112** and the second side **125** of the support layer **111** may be in contact across the entirety of their respective surface areas. The support layer **111** may be coupled to the second adhesive layer **112** by bonding, adhering, welding, or cross-linking, for example. In some embodiments, the plurality of apertures **132** may be distributed in the support layer **111** so that the plurality of apertures **132** are aligned with the plurality of apertures **116** of the second adhesive layer **112** if the support layer **111** and the second adhesive layer **112** are coupled. For example, each aperture **132** of the support layer **111** may be aligned with a corresponding aperture **116** of the second adhesive layer **112**.

(69) FIG. 4A is a schematic view of the assembled cover **106** in a first position on the epidermis, illustrating additional details that may be associated with some embodiments. The second side **120** of the film layer **114** may be coupled to first side **122** of the first adhesive layer **110**. The second side **124** of the first adhesive layer **110** may be coupled to the first side **123** of the support layer **111**. And the second side **125** of the support layer **111** may be coupled to the first side **126** of the second adhesive layer **112**. The cover **106** may be disposed over the epidermis to form the sealed therapeutic environment. The tackiness of the sealing adhesive of the second adhesive layer **112** may form sealing couplings **150** between the second side **128** of the second adhesive layer **112** and the epidermis to hold the cover **106** in an initial position as illustrated in FIG. 4A. The tackiness of the sealing adhesive of the second adhesive layer **112** may be such that the cover **106** may be removed and reapplied or repositioned. In some embodiments, the thickness **154** of the second adhesive layer **112** may prevent the first adhesive layer **110** from contacting the epidermis, forming a gap **139** between the first adhesive layer **110** and the epidermis.

(70) FIG. 4B is a schematic view of the cover **106** in a second position, illustrating additional details that may be associated with some embodiments. If the cover **106** is placed at a tissue site, a

force may be applied to the first side **118** of the film layer **114**. The force may cause at least some portion of the bonding adhesive of the first adhesive layer **110** to move through the apertures **116**. In response, the second side **124** of the first adhesive layer **110** may contact the epidermis and form bonding couplings **152**, as illustrated in the second position of FIG. **4B**. For example, the bonding couplings **152** may be located where the second side **124** of the first adhesive layer **110** adheres to the epidermis. In some embodiments, the bonding couplings **152** may have a peel force against the epidermis between about 0.5 N/25 mm and about 2 N/25 mm.

(71) The average effective diameter of the plurality of apertures **116** for the second adhesive layer **112** may be varied as one control of the tackiness or adhesion strength of the cover **106**. Generally, the strength of the bond of the bonding coupling **152** is proportional to the effective diameter of the plurality of apertures **116** of the second adhesive layer **112**, the thickness **154** of the second adhesive layer **112**, and the tackiness of the first adhesive layer **110**. The more of the first adhesive layer **110** that extends through the apertures **116**, the more bonding adhesive of the first adhesive layer **110** contacts the epidermis and the stronger the bond of the bonding coupling **152**. In some embodiments, the thickness **154** of the second adhesive layer **112** may permit more of the first adhesive layer **110** to extend through the apertures **116** and increase the bond of the bonding coupling **152**. As an example of the interplay, if a first bonding adhesive is used to form the first adhesive layer **110** and the thickness **154** of the second adhesive layer **112** is a first thickness, the average effective diameter of the plurality of apertures **116** may be a first effective diameter so that the bonding couplings **152** have a first bond strength. If the thickness **154** of the second adhesive layer **112** is increased to be larger than the first thickness, the average effective diameter may increase to be larger than the first effective diameter to achieve the first bond strength of the bonding coupling **152**. In some embodiments, the thickness **154** may be about 200 microns, the first adhesive layer **110** may be about 30 microns with a bonding adhesive having a tackiness of about 2000 g/25 cm wide strip, and the average effective diameter of each aperture **116** of the plurality of apertures **116** may be about 6 mm.

(72) The cover **106** can be manufactured in many ways including by following the operations provided below to produce the cover **106** without a scrim layer. The operations can be performed using manufacturing processes and equipment to manipulate the materials as described below. FIG. **5A** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **106**. In some embodiments, the support layer **111** may be provided and suitably disposed so that the first side **123** is supported by a suitable surface, and the second side **125** is exposed.

(73) FIG. **5B** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **106**. A sealing adhesive may be cast onto the second side **125** of the support layer **111**. In some embodiments, the sealing adhesive may be cured or otherwise cross-linked to form the second adhesive layer **112** and couple the second adhesive layer **112** to the second side **125** of the support layer **111**. In some embodiments, the sealing adhesive of the second adhesive layer **112** may be a two-part silicone system, such as those provided by Wacker, Dow, or NuSil. To cure, the two-part silicone system may be heated once applied to the second side **125** of the support layer **111**. Heating may encourage cross-linking or curing to cause the two-part silicone system to form a gel. In other embodiments, the sealing adhesive may be a one-part silicone system, and the sealing adhesive may be cured or cross-linked by exposing the one-part silicone system to ultraviolet light or ionizing radiation.

(74) In other embodiments, the support layer **111** may be laminated to the second adhesive layer **112**. For example, the sealing adhesive of the second adhesive layer **112** may be cast and cured as described above. After the sealing adhesive has cured to form the second adhesive layer **112**, the support layer **111** may be laminated to the second adhesive layer **112**.

(75) FIG. **5C** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **106**. The plurality of apertures **116** and the

plurality of apertures **132** may be formed following casting of the second adhesive layer **112** onto the support layer **111**. In some embodiments, the apertures **116** and the apertures **132** may be formed using shaped pins that puncture the support layer **111** and the second adhesive layer **112** as the support layer **111** and the second adhesive layer **112** move along a manufacturing path. In other embodiments, the apertures **116** and the apertures **132** may be formed by rolling a drum having shaped pins along the support layer **111** and the second adhesive layer **112**. The shaped pins may be configured to make the desired shape and size of the apertures **116** and the apertures **132**. In other embodiments, the apertures **116** and the apertures **132** may be cut or torn. For example, a punch may be used to puncture the support layer **111** and the second adhesive layer **112**. The apertures **116** and the apertures **132** may also be formed by melting portions of the support layer **111** and the second adhesive layer **112**. For example, a heated element, such as a poker may be applied to the support layer **111** and the second adhesive layer **112**, melting portions of the support layer **111** and the second adhesive layer **112**. In some embodiments, the apertures **116** and the apertures **132** may be formed in the second adhesive layer **112** and the support layer **111**, respectively, prior to the coupling of the support layer **111** and the second adhesive layer **112** to each other.

(76) In some embodiments, a mold may be used to form the second adhesive layer **112**. For example, the sealing adhesive may be cast into a mold having projections that extend into the sealing adhesive. After the sealing adhesive is cured, the mold and projections may be removed, leaving the apertures **116** formed in the second adhesive layer **112**. The support layer **111** may then be laminated to the second adhesive layer **112**. If the apertures **116** are formed with a mold, the apertures **132** may be separately formed in the support layer **111**.

(77) FIG. 5D is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **106**. In the example embodiment of FIG. 5D, the film layer **114** may be provided and positioned so that the first side **118** is supported by a suitable surface and the second side **120** is exposed.

(78) FIG. 5E is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **106**. For example, the first adhesive layer **110** may be cast onto the second side **120** of the film layer **114** of FIG. 5D. In other embodiments, the first adhesive layer **110** may be cast and cured, and the film layer **114** may then be laminated to the first adhesive layer **110**.

(79) FIG. 5F is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **106**. The support layer **111** and the second adhesive layer **112** may be positioned so that the first side **123** of the support layer **111** is proximate to the second side **124** of the first adhesive layer **110** of FIG. 5E. The first side **123** of the support layer **111** may be coupled to the second side **124** of the first adhesive layer **110** to form the cover **106** as shown in the example of FIG. 5F. In other embodiments, the second side **124** of the first adhesive layer **110** may be brought proximate to the first side **123** of the support layer **111**. The second side **124** of the first adhesive layer **110** may then be coupled to the first side **123** of the support layer **111**, forming the cover **106**.

(80) FIG. 6 is perspective view of a cover **206** partially assembled, which may be associated with some embodiments of the therapy system **100** of FIG. 1, for example. The cover **206** may be similar to the cover **106** as described above with respect to FIGS. 1-5F. Similar elements may have similar reference numbers that are indexed to **200**. As shown in FIG. 6, a film layer **214** may be coupled to a first adhesive layer **210**. For example, a second side **220** of the film layer **214** may be in contact with and coupled to a first side **222** of the first adhesive layer **210**. The film layer **214** and the first adhesive layer **210** may be coextensive. The film layer **214** may be coupled to the first adhesive layer **210** by bonding, adhering, welding, or cross-linking, for example.

(81) Similarly, a support layer **211** may be coupled to a second adhesive layer **212**. For example, a second side **225** of the support layer **211** may be in contact with and coupled to a first side **226** of the second adhesive layer **212**. The support layer **211** and the second adhesive layer **212** may be

coextensive. The support layer **211** may be coupled to the second adhesive layer **212** by bonding, adhering, welding, or cross-linking, for example. In some embodiments, a plurality of apertures **232** may be distributed in the support layer **211** so that the plurality of apertures **232** are aligned with a plurality of apertures **216** of the second adhesive layer **212**. The plurality of apertures **232** and the plurality of apertures **216** may be said to be coextensive so that each aperture **232** is aligned with a corresponding aperture **216**.

(82) In some embodiments, the support layer **211** may be removed prior to final assembly of the cover **206**. For example, as shown in FIG. 5, the support layer **211** may be removed by peeling the support layer **211** from the second adhesive layer **212**.

(83) FIG. 7 is a sectional view of the cover **206** illustrating additional details that may be associated with some embodiments. If the support layer **211** is removed, the first adhesive layer **210** may be coupled directly to the second adhesive layer **212** to produce the cover **206**. The cover **206** may be used as described above with respect to the cover **106** in FIG. 4A and FIG. 4B.

(84) The cover **206** can be manufactured without the use of a scrim layer. The operations can be performed using manufacturing processes and equipment to manipulate the materials as described below. FIG. 8A is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. In some embodiments, the support layer **211** may be provided and suitably disposed so that the first side **223** is supported by a suitable surface, and the second side **225** is exposed.

(85) FIG. 8B is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. A sealing adhesive may be cast onto the second side **225** of the support layer **211**. In some embodiments, the sealing adhesive may be cured or otherwise cross-linked to form the second adhesive layer **212** and couple the second adhesive layer **212** to the second side **225** of the support layer **211**. In some embodiments, the sealing adhesive of the second adhesive layer **212** may be a two-part silicone system, such as those provided by Wacker, Dow, or NuSil. To cure, the two-part silicone system may be heated once applied to the second side **225** of the support layer **211**. Heating may cause crosslinking or curing to cause the two-part silicone system to form a gel. In other embodiments, the sealing adhesive may be a one-part silicone system, and the sealing adhesive may be cured or crosslinked by exposing the one-part silicone system to ultraviolet light or ionizing radiation.

(86) In other embodiments, the support layer **211** may be laminated to the second adhesive layer **212**. For example, the sealing adhesive of the second adhesive layer **212** may be cast and cured as described above. After the sealing adhesive has cured to form the second adhesive layer **212**, the support layer **211** may be laminated to the second adhesive layer **212**.

(87) FIG. 8C is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. The plurality of apertures **216** and the plurality of apertures **232** may be formed following casting of the second adhesive layer **212** onto the support layer **211**. In some embodiments, the apertures **216** and the apertures **232** may be formed using shaped pins that puncture the support layer **211** and the second adhesive layer **212** as the support layer **211** and the second adhesive layer **212** move along a manufacturing path. In other embodiments, the apertures **216** and the apertures **232** may be formed by rolling a drum having shaped pins along the support layer **211** and the second adhesive layer **212**. The shaped pins may be configured to make the desired shape and size of the apertures **216** and the apertures **232**. In other embodiments, the apertures **216** and the apertures **232** may be cut or torn. For example, a punch may be used to puncture the support layer **211** and the second adhesive layer **212**. The apertures **216** and the apertures **232** may also be formed by melting portions of the support layer **211** and the second adhesive layer **212**. For example, a heated element, such as a poker may be applied to the support layer **211** and the second adhesive layer **212**, melting portions of the support layer **211** and the second adhesive layer **212**. In some embodiments, the apertures **216** and the apertures **232** may be formed in the second adhesive layer **212** and the support layer **211**, respectively, prior to the

coupling of the support layer **211** and the second adhesive layer **212**.

(88) In some embodiments, a mold may be used to form the second adhesive layer **212**. For example, the sealing adhesive may be cast into a mold having projections that extend into the sealing adhesive. After the sealing adhesive is cured, the mold and projections may be removed, leaving the apertures **216** formed in the second adhesive layer **212**. The support layer **211** may then be laminated to the second adhesive layer **212**. In some embodiments, the support layer **211** may not include the plurality of apertures **232**.

(89) FIG. **8D** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. For example, the film layer **214** may be provided and positioned so that the first side **218** is supported by a suitable surface and the second side **220** is exposed.

(90) FIG. **8E** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. As illustrated in the example of FIG. **8E**, the first adhesive layer **210** may be cast onto the second side **220** of the film layer **214** of FIG. **8D**. In other embodiments, the first adhesive layer **210** may be cast and cured and the film layer **214** may then be laminated to the first adhesive layer **210**.

(91) FIG. **8F** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. Following a curing or crosslinking process, the second adhesive layer **212** and the support layer **211** may be positioned so that the first side **223** of the support layer **211** is in contact with a surface, leaving the second side **228** of the second adhesive layer **212** free. The second side **224** of the first adhesive layer **210** of FIG. **8E** may be brought proximate to the second side **228** of the second adhesive layer **212**. The second side **224** of the first adhesive layer **210** may then be coupled to the second side **228** of the second adhesive layer **212**, as illustrated in the example of FIG. **8F**, forming one embodiment of the cover **206**.

(92) FIG. **8G** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **206**. After coupling of the second side **224** of the first adhesive layer **210** to the second side **228** of the second adhesive layer **212** in the example of FIG. **8F**, the cover **206** may be repositioned so that the first side **223** of the support layer **211** is exposed. The support layer **211** may then be removed from the first side **226** of the second adhesive layer **212**.

(93) FIG. **9A** is a perspective view of the cover **206**, illustrating additional details that may be associated with some embodiments. The cover **206** includes the film layer **214**, the first adhesive layer **210** coupled to the film layer **214**, and the second adhesive layer **212** having the apertures **216** coupled to the first adhesive layer **210**. In some embodiments, the cover **206** may include one or more sizing lines **240**. Each sizing line **240** may extend across the cover **206** from a first edge **244** to a second edge **246**. In some embodiments, the sizing lines **240** may be equidistantly spaced across the cover **206**. For example, each sizing line **240** may be between about 5 centimeters and about 30 centimeters from adjacent sizing lines **240**. In other embodiments, each sizing line **240** may be about 28 centimeters from adjacent sizing lines **240**.

(94) A sizing line **240** may be a weakened portion of the cover **206** that facilitates separation of a first portion of the cover **206** from a second portion of the cover **206**. In some embodiments, each sizing line **240** may be formed by perforating the cover **206**. For example, each sizing line **240** may have a plurality of perforations **242**. In some embodiments, each perforation **242** may be between about 25 mm and about 60 mm from adjacent perforations **242** in the same sizing line **240**. While each sizing line **240** is shown in portions of the cover **206** between apertures **216**, in other embodiments, the sizing lines **240** may pass through the apertures **216**.

(95) FIG. **9B** is a schematic view of the cover **206**, illustrating additional details that may be associated with some embodiments of the perforations **242**. As shown in FIG. **9B**, each perforation **242** may be formed in the cover **206** that extends through the film layer **214**, the first adhesive layer **210**, and the second adhesive layer **212**. If the cover **206** is applied to a tissue site and a sizing line

240 extends across the sealed therapeutic environment, the sealing adhesive of the second adhesive layer **212** may expand and fill the perforation **242** to seal the cover **206**.

(96) FIG. **9C** is a sectional view of the cover **206**, illustrating additional details that may be associated with some embodiments of the perforations **242**. As shown in FIG. **9C**, each perforation **242** may be formed in the cover **206** that extends through the film layer **214** and the first adhesive layer **210**, but not the second adhesive layer **212**. The cover **206** may be torn along the sizing line **240** to separate portions of the cover **206**.

(97) FIG. **10** is a perspective exploded view of an example embodiment of a cover **306** that can be used with the therapy system **100**. The cover **306** may include a first adhesive layer **310**, a support layer **311**, a second adhesive layer **312**, and a film layer **314**. The first adhesive layer **310**, the support layer **311**, the second adhesive layer **312**, and the film layer **314** may be similar to and operate as described above with respect to the first adhesive layer **110**, the support layer **111**, the second adhesive layer **112**, and the film layer **114**.

(98) The film layer **314** may have a first side **318** and a second side **320**. The first adhesive layer **310** may also have a first side **322** and a second side **324**. The first side **322** of the first adhesive layer **310** may face the second side **320** of the film layer **314**. The first adhesive layer **310** may have a thickness **356**, and in some embodiments, the thickness **356** of the first adhesive layer **310** may be about 30 microns to about 60 microns. In some embodiments, the first adhesive layer **310** may be coupled to the film layer **314**.

(99) The support layer **311** may have a first side **323** and a second side **325**. In some embodiments, the support layer **311** may have a plurality of apertures **332**. Each aperture **332** of the plurality of apertures **332** may extend through the support layer **311** from the first side **323** to the second side **325**. In some embodiments, the support layer **311** may have a thickness of about 15 microns. The first side **323** of the support layer **311** may also have a substantially smooth surface.

(100) The cover **306** may also include a third adhesive layer **313**. The third adhesive layer **313** may be disposed between the support layer **311** and the second adhesive layer **312**. The third adhesive layer **313** may have a first side **334** and a second side **336**. The first side **334** of the third adhesive layer **313** may face the second side **325** of the support layer **311**. In some embodiments, the third adhesive layer **313** may have a plurality of apertures **338** that extend through the third adhesive layer **313** from the first side **334** to the second side **336**. The third adhesive layer **313** may be an acrylic adhesive having a coating weight between 25-65 grams per square meter (g.s.m.). In some embodiments, the third adhesive layer **313** may have a thickness **358**. The thickness **358** of the third adhesive layer **313** may be between about 30 microns and about 60 microns. In other embodiments, the thickness **358** of the third adhesive layer **313** may be between about 15 microns and about 25 microns. Thicker adhesives, or combinations of adhesives, may be applied in some embodiments to improve the seal and reduce leaks. Other example embodiments of the third adhesive layer **313** may include a double-sided tape, paste, hydrocolloid, hydrogel, silicone gel, or organogel.

(101) The second adhesive layer **312** may have a first side **326** and a second side **328**. The first side **326** of the second adhesive layer **312** may face the second side **336** of the third adhesive layer **313** and have a thickness **354**. In some embodiments, the second adhesive layer **312** may have a plurality of apertures **316** that extend through the second adhesive layer **312** from the first side **326** to the second side **328**.

(102) The plurality of apertures **332**, **338**, **316** may be numerous shapes, including without limitation, circles, squares, stars, ovals, polygons, slits complex curves, rectilinear shapes, or triangles. Each aperture **332**, **338**, **316** of the plurality of apertures **332**, **338**, **316** may have an effective diameter. The average effective diameter is typically in the range of about 4 mm to about 50 mm. The plurality of apertures **332**, **338**, **316** may have a uniform pattern or may be randomly distributed on the support layer **311**, the third adhesive layer **313**, and the second adhesive layer **312**, respectively. In some embodiments, the apertures **332**, **338**, **316** may be distributed so that

centers of adjacent apertures **332**, **338**, **316** may be separated by between about 5 mm and about 100 mm. In other embodiments, the apertures **332**, **338**, **316** may be distributed so that the centers of adjacent apertures **332**, **338**, **316** may be separated by about 10 mm.

(103) Regardless of the shape of the plurality of apertures **332**, **338**, **316**, the plurality of apertures **332**, **338**, **316** in the support layer **311**, the third adhesive layer **313**, and the second adhesive layer **312**, respectively, may leave void spaces in the support layer **311**, the third adhesive layer **313**, and the second adhesive layer **312**, respectively. The percentage of void space of the plurality of apertures **332**, **338**, **316** may be equal to the percentage of the volume or surface area of the void spaces created by the plurality of apertures **332**, **338**, **316** to the total volume or surface area of the support layer **311**, the third adhesive layer **313**, and the second adhesive layer **312**, respectively. In some embodiments, the percentage of void space may be between about 40% and about 75%. In other embodiments, the percentage of void space may be about 55%. The organization of the plurality of apertures **332**, **338**, **316** can also impact the percentage of void space. The plurality of apertures **332**, **338**, **316** may be formed by punching, cutting, melting, or drilling the support layer **311**, the third adhesive layer **313**, and the second adhesive layer **312**, respectively.

(104) To manufacture the cover **306**, the third adhesive layer **313** may be coupled to the support layer **311**, and the second adhesive layer **312** may be coupled to the third adhesive layer **313** to form a first assembly. The apertures **332**, **338**, and **316** may be formed in the first assembly. The first adhesive layer **310** may be coupled to the film layer **314** to form a second assembly. The first assembly and the second assembly may be coupled to each other so that the support layer **311** is coupled to the first adhesive layer **310**.

(105) In some embodiments, the first adhesive layer **310** and the third adhesive layer **313** may be formed from a material having a substantially similar composition. For example, the first adhesive layer **310** and the third adhesive layer **313** may be formed from the same acrylic adhesive. If the first adhesive layer **310** and the third adhesive layer **313** are formed from the same acrylic adhesive, the first adhesive layer **310** and the third adhesive layer **313** may react similarly in response to outside stimuli, such as the passage of time and exposure to heat. In some embodiments, the first adhesive layer **310** and the third adhesive layer **313** may expand and/or contract at substantially the same rate in response to the same stimuli. By forming the first adhesive layer **310** and the third adhesive layer **313** from the same material, the cover **306** can reduce instances of acrylic adhesive exposure about a perimeter of the cover **306** compared to covers using dissimilar acrylic adhesives to form the first adhesive layer **310** and the third adhesive layer **313**.

(106) In some embodiments, the first side **323** or the second side **325** of the support layer **311** may include printing. For example, the first side **323** of the support layer **311** may be marked with a manufacturer's brand or trademark. In other examples, information regarding suggested uses for the cover **306** may be printed on the first side **323** of the support layer **311**. The printing may be visible through the first adhesive layer **310** and the film layer **314** during use of the cover **306**. Because the support layer **311** is covered by the first adhesive layer **310** and the film layer **314**, the printing is less susceptible to washing off during use than if the printing is applied to the film layer **314**. For example, if fluid, including liquid, comes into contact with the film layer **314**, it will not come into contact with the support layer **311** or printing formed on the support layer **311**.

(107) FIG. **11** is a perspective exploded view of an example embodiment of a cover **406** that can be used with the therapy system **100**. The cover **406** may include a first adhesive layer **410**, a support layer **411**, a second adhesive layer **412**, a third adhesive layer **413**, and a film layer **414**. The first adhesive layer **410**, the support layer **411**, the second adhesive layer **412**, and the film layer **414** may be similar to and operate as described above with respect to the first adhesive layer **110**, the support layer **111**, the second adhesive layer **112**, and the film layer **114**.

(108) The film layer **414** may have a first side **418** and a second side **420**. The first adhesive layer **410** may also have a first side **422** and a second side **424**. The first side **422** of the first adhesive layer **410** may face the second side **420** of the film layer **414**. The first adhesive layer **410** may

have a thickness **456**, and in some embodiments, the thickness **456** of the first adhesive layer **410** may be between about 40 microns and about 60 microns. In some embodiments, the first adhesive layer **410** may be coupled to the film layer **414**.

(109) The support layer **411** may have a first side **423** and a second side **425**. In some embodiments, the support layer **411** may have a plurality of apertures **432**. Each aperture **432** of the plurality of apertures **432** may extend through the support layer **411** from the first side **423** to the second side **425**. The third adhesive layer **413** may be disposed between the support layer **411** and the first adhesive layer **410**. The third adhesive layer **413** may have a first side **434** and a second side **436**. The first side **434** of the third adhesive layer **413** may face the second side **424** of the first adhesive layer **410**. The second side **436** of the third adhesive layer **413** may face first side **423** of the support layer **411**. In some embodiments, the third adhesive layer **413** may have a plurality of apertures **438** that extend through the third adhesive layer **413** from the first side **434** to the second side **436**. The third adhesive layer **413** may be an acrylic adhesive having a coating weight between 25-65 grams per square meter (g.s.m.). In some embodiments, the third adhesive layer **413** may have a thickness **458**. The thickness **458** of the third adhesive layer **413** may be between about 15 microns and about 25 microns. Thicker adhesives, or combinations of adhesives, may be applied in some embodiments to improve the seal and reduce leaks. Other example embodiments of the third adhesive layer **413** may include a double-sided tape, paste, hydrocolloid, hydrogel, silicone gel, or organogel.

(110) The second adhesive layer **412** may have a first side **426** and a second side **428**. The first side **426** of the second adhesive layer **412** may face the second side **425** of the support layer **411** and have a thickness **454**. In some embodiments, the second adhesive layer **412** may have a plurality of apertures **416** that extend through the second adhesive layer **412** from the first side **426** to the second side **428**.

(111) The plurality of apertures **438**, **432**, **416** may be numerous shapes, including without limitation, circles, squares, stars, ovals, polygons, slits complex curves, rectilinear shapes, or triangles. Each aperture **438**, **432**, **416** of the plurality of apertures **438**, **432**, **416** may have an effective diameter. The average effective diameter is typically in the range of about 4 mm to about 50 mm. The plurality of apertures **438**, **432**, **416** may have a uniform pattern or may be randomly distributed on the support layer **411**, the third adhesive layer **413**, and the second adhesive layer **412**, respectively. In some embodiments, the apertures **438**, **432**, **416** may be distributed so that centers of adjacent apertures **438**, **432**, **416** may be separated by between about 5 mm and about 100 mm. In other embodiments, the apertures **438**, **432**, **416** may be distributed so that the centers of adjacent apertures **438**, **432**, **416** may be separated by about 10 mm.

(112) Regardless of the shape of the plurality of apertures **438**, **432**, **416**, the plurality of apertures **438**, **432**, **416** in the third adhesive layer **413**, the support layer **411**, and the second adhesive layer **412**, respectively, may leave void spaces in the third adhesive layer **413**, the support layer **411**, and the second adhesive layer **412**, respectively. The percentage of void space of the plurality of apertures **438**, **432**, **416** may be equal to the percentage of the volume or surface area of the void spaces created by the plurality of apertures **438**, **432**, **416** to the total volume or surface area of the third adhesive layer **413**, the support layer **411**, and the second adhesive layer **412**, respectively. In some embodiments, the percentage of void space may be between about 40% and about 75%. In other embodiments, the percentage of void space may be about 55%. The organization of the plurality of apertures **438**, **432**, **416** can also impact the percentage of void space. The plurality of apertures **438**, **432**, **416** may be formed by punching, cutting, melting, or drilling the third adhesive layer **413**, the support layer **411**, and the second adhesive layer **412**, respectively.

(113) FIG. 12A is a schematic view of the assembled cover **406** in a first position on the epidermis, illustrating additional details that may be associated with some embodiments. The second side **420** of the film layer **414** may be coupled to first side **422** of the first adhesive layer **410**. The second side **424** of the first adhesive layer **410** may be coupled to the first side **434** of the third adhesive

layer **413**. The second side **436** of the third adhesive layer **413** may be coupled to the first side **423** of the support layer **411**. And the second side **425** of the support layer **411** may be coupled to the first side **426** of the second adhesive layer **412**. The cover **406** may be disposed over the epidermis to form the sealed therapeutic environment. The tackiness of the sealing adhesive of the second adhesive layer **412** may form sealing couplings **450** between the second side **428** of the second adhesive layer **412** and the epidermis to hold the cover **406** in an initial position as illustrated in FIG. **12A**. The tackiness of the sealing adhesive of the second adhesive layer **412** may be such that the cover **406** may be removed and reapplied or repositioned. In some embodiments, the thickness **454** of the second adhesive layer **412** may prevent the first adhesive layer **410** from contacting the epidermis, forming a gap **439** between the first adhesive layer **410** and the epidermis. In some embodiments, sidewalls of the gap **439** comprise the aperture **416** of the second adhesive layer **412**, the aperture **432** of the support layer **411**, and the aperture **438** of the third adhesive layer **413**.

(114) FIG. **12B** is a schematic view of the cover **406** in a second position, illustrating additional details that may be associated with some embodiments. If the cover **406** is placed at a tissue site, a force may be applied to the first side **418** of the film layer **414**. The force may cause at least some portion of the bonding adhesive of the first adhesive layer **410** to move through the apertures **438**, the apertures **432**, and the apertures **416**. In response, the second side **424** of the first adhesive layer **410** may contact the epidermis and form bonding couplings **452**, as illustrated in the second position of FIG. **12B**. For example, the bonding couplings **452** may be located where the second side **424** of the first adhesive layer **410** adheres to the epidermis. In some embodiments, the bonding couplings **452** may have a peel force against the epidermis between about 0.5 N/25 mm and about 9.5 N/25 mm.

(115) The average effective diameter of the apertures **438**, the apertures **432**, and the apertures **416** may be varied as one control of the tackiness or adhesion strength of the cover **406**. Generally, the strength of the bond of the bonding coupling **452** is proportional to the effective diameter of the apertures **438**, the apertures **432**, and the apertures **416**; the thickness **454** of the second adhesive layer **412**, the thickness of the support layer **411**, and the thickness **458** of the third adhesive layer **413**; and the tackiness of the first adhesive layer **410**. The more of the first adhesive layer **410** that extends through the across the gap **439**, the more bonding adhesive of the first adhesive layer **410** contacts the epidermis and the stronger the bond of the bonding coupling **452**. In some embodiments, the thickness **454** of the second adhesive layer **412**, the thickness of the support layer **411**, and the thickness **458** of the third adhesive layer **413** may permit more of the first adhesive layer **410** to extend across the gap **439** and increase the bond of the bonding coupling **452**.

(116) As an example of the interplay, if a first bonding adhesive is used to form the first adhesive layer **410** and the thickness **454** of the second adhesive layer **412**, the support layer **411**, and the thickness **458** of the third adhesive layer **413** is collectively a first thickness, the average effective diameter of the apertures **438**, the apertures **432**, and the apertures **416** may be a first effective diameter so that the bonding couplings **452** have a first bond strength. If the thickness **454** of the second adhesive layer **412**, the thickness of the support layer **411**, and the thickness **458** of the third adhesive layer **413** is collectively increased to be larger than the first thickness, the average effective diameter of the apertures **438**, the apertures **432**, and the apertures **416** may increase to be larger than the first effective diameter to achieve the first bond strength of the bonding coupling **452**. In some embodiments, the thickness **454** may be about 250 microns, the first adhesive layer **410** may be about 30 microns with a bonding adhesive having a tackiness of about 9.5 N/25 mm wide strip, and the average effective diameter of each aperture **416** of the plurality of apertures **416** may be about 7 mm.

(117) The cover **406** can be manufactured in many ways including by following the operations provided below to produce the cover **406** without a scrim layer. The operations can be performed using manufacturing processes and equipment to manipulate the materials as describe below. FIG. **13A** is a schematic view illustrating additional details that may be associated with some

embodiments of the manufacturing process of the cover **406**. In some embodiments, the support layer **411** may be provided and suitably disposed so that the second side **425** is supported by a suitable surface, and the first side **423** is exposed.

(118) FIG. **13B** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **406**. The third adhesive layer **413** may be cast onto the first side **423** of the support layer **411**. In other embodiments, the third adhesive layer **413** may be cast and cured, and the support layer **411** may be laminated to the third adhesive layer **413**.

(119) FIG. **13C** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **406**. The support layer **411** and the third adhesive layer **413** may be oriented so that the first side **434** of the third adhesive layer **413** is supported by a suitable surface, and the second side **425** of the support layer **411** is exposed. A sealing adhesive may be cast onto the second side **425** of the support layer **411**. In some embodiments, the sealing adhesive may be cured or otherwise cross-linked to form the second adhesive layer **412** and couple the second adhesive layer **412** to the second side **425** of the support layer **411**. In some embodiments, the sealing adhesive of the second adhesive layer **412** may be a two-part silicone system, such as those provided by Wacker, Dow, or NuSil. To cure, the two-part silicone system may be heated once applied to the second side **425** of the support layer **411**. Heating may cause cross-linking or curing to cause the two-part silicone system to form a gel. In other embodiments, the sealing adhesive may be a one-part silicone system, and the sealing adhesive may be cured or cross-linked by exposing the one-part silicone system to ultraviolet light or ionizing radiation.

(120) In other embodiments, the support layer **411** may be laminated to the second adhesive layer **412**. For example, the sealing adhesive of the second adhesive layer **412** may be cast and cured as described above. After the sealing adhesive has cured to form the second adhesive layer **412**, the support layer **411** may be laminated to the second adhesive layer **412**.

(121) FIG. **13D** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **406**. The plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438** may be formed following casting of the second adhesive layer **412** and the third adhesive layer **413** onto the support layer **411**. In some embodiments, the plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438** may be formed using shaped pins that puncture the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413** as the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413** move along a manufacturing path. In other embodiments, the plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438** may be formed by rolling a drum having shaped pins along the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413**. The shaped pins may be configured to make the desired shape and size of the plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438**. In other embodiments, the plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438** may be cut or torn. For example, a punch may be used to puncture the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413**. The plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438** may also be formed by melting portions of the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413**. For example, a heated element, such as a poker may be applied to the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413**, melting portions of the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413**. In some embodiments, the plurality of apertures **416**, the plurality of apertures **432**, and the plurality of apertures **438** may be formed in the second adhesive layer **412**, the support layer **411**, and the third adhesive layer **413**, respectively, prior to the coupling of the support layer **411**, the second adhesive layer **412**, and the third adhesive layer **413** to each other.

(122) In some embodiments, a mold may be used to form the second adhesive layer **412**. For example, the sealing adhesive may be cast into a mold having projections that extend into the sealing adhesive. After the sealing adhesive is cured, the mold and projections may be removed, leaving the apertures **416** formed in the second adhesive layer **412**. The support layer **411** may then be laminated to the second adhesive layer **412**. The third adhesive layer **413** may then be cast onto the first side **423** of the support layer **411**. If the apertures **416** are formed with a mold, the apertures **432** and the apertures **438** may be separately formed in the support layer **411** and the third adhesive layer **413**, respectively.

(123) FIG. **13E** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **406**. In the example embodiment of FIG. **13E**, the film layer **414** may be provided and positioned so that the first side **418** is supported by a suitable surface and the second side **420** is exposed.

(124) FIG. **13F** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **406**. For example, the first adhesive layer **410** may be cast onto the second side **420** of the film layer **414** of FIG. **13E**. In other embodiments, the first adhesive layer **410** may be cast and cured, and the film layer **414** may then be laminated to the first adhesive layer **410**.

(125) FIG. **13G** is a schematic view illustrating additional details that may be associated with some embodiments of the manufacturing process of the cover **406**. The support layer **411** and the second adhesive layer **412** may be positioned so that the first side **434** of the third adhesive layer **413** is proximate to the second side **424** of the first adhesive layer **410** of FIG. **13E**. The first side **434** of the third adhesive layer **413** may be coupled to the second side **424** of the first adhesive layer **410** to form the cover **406** as shown in the example of FIG. **13G**. In other embodiments, the second side **424** of the first adhesive layer **410** may be brought proximate to the first side **434** of the third adhesive layer **413**. The second side **424** of the first adhesive layer **410** may then be coupled to the first side **434** of the third adhesive layer **413**, forming the cover **406**.

(126) FIG. **14** is a schematic view of the cover **406** illustrating additional details that may be associated with some embodiments. The cover **406** may be substantially transparent. For example, the cover **406** may have a low absorbance and transmit light through the cover **406** at a high rate. As shown in FIG. **14**, light, as represented by arrows **440**, may pass through the cover **406** unobstructed, having no physical barrier to prevent light penetration. As a result, light **440** may be reflected by objects to which the cover **406** is attached.

(127) In some embodiments, the cover **106**, the cover **206**, the cover **306**, and the cover **406** may be substantially transparent. For example, in some embodiments the transmittance of the cover **106**, the cover **206**, the cover **306**, and the cover **406** for visible electro-magnetic radiation may be between about 70% and about 100%. In other embodiments, the transmittance of the cover **106**, the cover **206**, the cover **306**, and the cover **406** for visible electro-magnetic radiation may be about 90%. Generally, transmittance may refer to the ratio of the amount of radiation passing through a surface to the amount of radiation falling on the surface. Materials and objects having a high transmittance, for example, approaching 1 or 100%, may be transparent. Factors that may impact the transmittance of the cover **106**, the cover **206**, the cover **306**, and the cover **406** may include the amount of electro-magnetic radiation the materials of the cover **106**, the cover **206**, the cover **306**, and the cover **406** absorb, and the amount of the electro-magnetic radiation the materials of the cover **106**, the cover **206**, the cover **306**, and the cover **406** scatter. By manufacturing the cover **106**, the cover **206**, the cover **306**, and the cover **406** without a scrim layer, the amount of electro-magnetic radiation the materials of the cover absorb and scatter can be decreased compared to a cover having a scrim layer. The tendency of a material to absorb, scatter, or otherwise attenuate light can be referred to as absorbance. Manufacturing the cover **106**, the cover **206**, the cover **306**, and the cover **406** without a scrim layer can increase the ability of the cover **106**, the cover **206**, the cover **306**, and the cover **406** to transmit electro-magnetic radiation, i.e., decrease absorbance. For

example, referring to FIG. 1, the transmittance of the cover **106** may permit the tissue interface **108** and the tissue site to be seen through the cover **106**.

(128) FIG. **15** is a graphical depiction of absorbance measured across covers, illustrating additional details that may be associated with some embodiments. In the example, covers having a scrim layer and covers having no scrim layer, such as the cover **406**, were tested for absorbance. The testing apparatus included a substantially transparent surface having a light sensor on a first side and a light source on a second side. The light source was configured to produce a substantially white light, and the light sensor was configured to measure the transmittance through the transparent surface. The exemplary cover was placed on the second side of the transparent surface. The light source was turned on, and the transmittance through the cover was measured by the light sensor. Based on the measured transmittance, the absorbance was determined. Absorbance is a dimensionless measurement that reflects the amount of light absorbed and attenuated, which can include both light reflected and scattered. Generally, a higher absorbance indicates greater opaqueness of an object and lower absorbance indicates greater transparency.

(129) A cover **501** having a polyurethane scrim layer formed from a silicone gel was tested for absorbance. A first side of the scrim layer was coated with a silicone adhesive, and a second side of the scrim layer was coated with an acrylic adhesive. The scrim layer, silicone adhesive, and acrylic adhesive was perforated. The perforated scrim layer, silicone adhesive, and acrylic adhesive was laminated to a polyurethane film also having an acrylic adhesive. The acrylic adhesive of the scrim layer was adhered to the acrylic adhesive of the polyurethane film. The polyurethane film had a thickness of about 40 microns, the acrylic adhesive had a thickness of about 50 microns, and the scrim layer had a thickness of about 50 microns. A coating weight of the silicone adhesive for the cover **501** was about 250 gsm. As shown in FIG. **15**, the absorbance of the cover **501** showed variation for wavelengths less than about 300 nm. In the visible light spectrum between about 390 nm and about 700 nm, the absorbance of the cover **501** was about 1. An absorbance of about 1 indicates that approximately 10% of the light is transmitted through the cover **501** and about 90% of the light is attenuated by the cover **501**. The cover **501** may be considered substantially opaque.

(130) Another cover **502** was also tested for absorbance. The cover **502** was constructed as described above with respect to the cover **501**. The cover **502** included a thicker silicone adhesive coated onto the scrim layer. For the cover **502**, the coating weight of the silicone adhesive was about 500 gsm. As shown in FIG. **15**, the cover **502** has an absorbance in the visible light spectrum (between about 390 nm and about 700 nm) of about 1.3, indicating that the cover **502** permitted less light transmission than the cover **501**. The cover **502** may be considered substantially opaque.

(131) Another cover **503** was also tested for absorbance. The cover **503** was constructed as described above with respect to the cover **406**. For the cover **503**, the film layer **414** had a thickness of about 25 microns, the first adhesive layer **410** had a coating weight of about 45 gsm, the third adhesive layer **413** had a coating weight of about 20 gsm, the support layer **411** had a thickness of about 15 microns, and the second adhesive layer **412** had a coating weight of about 150 gsm. As shown in FIG. **15**, the cover **503** has an absorbance less than 1, and very close to 0, indicating that greater than about 80% of the light transmitted through the cover **503**. In some embodiments, the absorbance of the cover **503** was less than about 0.1. The average absorbance of the cover **503** across the visible light spectrum between about 390 nm and about 700 nm was about 0.02.

Similarly, a cover **504** was tested. The cover **504** is similar to the cover **503** as described above. For the cover **504**, the second adhesive layer **412** had a coating weight of about 300 gsm. As shown in FIG. **15**, the absorbance of the cover **504** is slightly greater than the cover **503**. The average absorbance of the cover **504** across the visible spectrum from 390 nm to about 700 nm is about 0.07. Both the cover **503** and the cover **504** may be considered substantially transparent.

(132) The cover **503** and the cover **504**, produced as described above with respect to the cover **406**, have substantially lower absorbance than covers, such as the cover **501** and the cover **502**, that include a scrim layer. As a result, the cover **503** and the cover **504** are substantially transparent,

permitting light to pass through the cover **503** and the cover **504** to be reflected by objects underneath the cover **503** and the cover **504**. The transparency of the cover **503** and the cover **504** can permit a clinician to see color and other details of a tissue site. Color that can be perceived through an object, such as glass, is affected by the wavelengths of light that are absorbed and reflected by the object. For example, an object that appears visibly red may reflect electromagnetic radiation having wavelengths in the red spectrum while absorbing electromagnetic radiation having wavelengths of the non-red spectrum. In some embodiments, the absorbance of the cover **503** and the cover **504** is low, less than about 0.07, permitting a substantial portion of the electro-magnetic radiation in the visible spectrum to pass through the cover **503** and the cover **504**. Consequently, the objects perceived through the cover **503** and the cover **504**, such as a tissue site or the tissue interface **108**, may have a same appearance as the object perceived independently of the cover **503** and the cover **504**, i.e., as if nothing is between the clinician and the tissue site.

(133) FIG. **16** is a graphical depiction of leak rates for exemplary covers, illustrating additional details that may be associated with some embodiments. In the example, each cover was positioned on a testing apparatus having a grid of channels formed therein. The channels of the grid had a depth of approximately 150 microns, and each channel was spaced approximately 0.75 mm from adjacent channels. The covers were placed over the grid, and sealed to the surface of the grid. A negative-pressure therapy apparatus was fluidly coupled to an area between the cover and the grid, and negative-pressure therapy was conducted. During the test, a negative pressure of about 125 mmHg was maintained in the area between the cover and the grid. A leak rate of the cover was then calculated. The leak rate was determined based on the operation of the negative-pressure therapy apparatus to maintain the area under the cover at approximately 125 mmHg negative pressure. For example, the operation of the negative-pressure therapy apparatus was monitored. Based on a known flow rate for the negative-pressure therapy apparatus, the operational time of the negative-pressure therapy apparatus was measured to determine a total volume of fluid moved by the negative-pressure therapy apparatus during the operational time. From the total volume of fluid moved by the negative-pressure therapy apparatus for the operational time, the leak rate was calculated.

(134) A first cover **601** was tested as described above to determine its leak rate. The cover **601** is similar to and includes the components of the cover **406**. For the cover **601**, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having a thickness of about 40 microns. The second adhesive layer **412** was formed from a silicone gel, such as a 6054 silicone gel produced by Scapa™, having a coating weight of about 150 gsm. And the third adhesive layer was formed from an acrylic adhesive having a coating weight of about 20 gsm. As shown in FIG. **16**, the cover **601** exhibited an increasing leak rate over time. The initial leak rate was about 0 cubic centimeters per minute (ccm), and the leak rate increased to about 1600 ccm after approximately 90 minutes.

(135) A cover **602** was also tested as described above to determine its leak rate. The cover **602** is similar to and includes the components of the cover **406**. For the cover **602**, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having a thickness of about 25 microns. The second adhesive layer **412** comprised a silicone gel, such as a RX1423s silicone gel produced by Scapa™, having a coating weight of about 150 gsm. The third adhesive layer comprises an acrylic adhesive having a coating weight of about 20 gsm. As shown in FIG. **16**, the cover **602** maintained a low leak rate over time. For example, the initial leak rate was slightly greater than about 0 ccm, averaged about 23 ccm, and the leak rate was maintained for greater than 120 minutes.

(136) A cover **603** was also tested as described above to determine its leak rate. The cover **603** is similar to and includes components of the cover **406**; however, the cover **603** did not include the third adhesive layer **413**. In the example, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having thickness of about 25 microns. The second adhesive layer **412** was formed from a polyurethane gel, such as a 9772B polyurethane gel produced by Scapa™ having a coating weight of about 200 gsm. As shown in FIG. **16**, the cover **603** maintained a low leak rate over time as shown by line **603**. The initial leak rate was slightly greater than the cover **602**, averaged about 33 ccm, and the leak rate was maintained for greater than 120 minutes.

(137) Each of the cover **601**, the cover **602**, and the cover **603**, was constructed without a scrim layer. The cover **601**, the cover **602**, and the cover **603** each exhibited the ability to provide a good seal over a tissue site. Consequently, the construction of the cover **601**, the cover **602**, and the cover **603**, having a film layer, a first adhesive layer, a third adhesive layer, a support layer, and a second adhesive layer, produces a cover suitable for use with negative-pressure therapy while increasing the ability to monitor a tissue site visually without having to remove the cover.

(138) FIG. **17** is a graphical depiction of leak rates for exemplary covers, illustrating additional details that may be associated with some embodiments. In the example, each cover was positioned on a testing apparatus having a grid of channels formed therein. The channels of the grid had a depth of approximately 150 microns, and each channel was spaced approximately 0.75 mm from adjacent channels. The cover was placed over the grid, and sealed to the surface of the grid. A secondary cover having an identical configuration was then placed over edges of the underlying cover. A negative-pressure therapy apparatus was fluidly coupled to an area between the cover and the grid, and negative-pressure therapy was conducted. During the test, a negative pressure of about 125 mmHg was maintained in the area between the cover and the grid. A leak rate of the covers was then calculated. The leak rate was determined based on the operation of the negative-pressure therapy apparatus to maintain the area under the exemplary covers at approximately 125 mmHg negative pressure. For example, the operation of the negative-pressure therapy apparatus was monitored. Based on a known flow rate for the negative-pressure therapy apparatus, the operational time of the negative-pressure therapy apparatus was measured to determine a total volume of fluid moved by the negative-pressure therapy apparatus during the operational time. From the total volume of fluid moved by the negative-pressure therapy apparatus for the operational time, the leak rate was calculated.

(139) A first cover **701** was tested as described above to determine its leak rate. The cover **701** is similar to and includes the components of the cover **406**. For the cover **701**, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having a thickness of about 40 microns. The second adhesive layer **412** was formed from a silicone gel, such as a 6054 silicone gel produced by Scapa™, having a coating weight of about 150 gsm. The third adhesive layer was formed from an acrylic adhesive having a coating weight of about 20 gsm. As shown in FIG. **17**, the cover **701** exhibited an increasing leak rate over time. The initial leak rate was about 0 ccm, and the leak rate varied over time from 0 minutes to approximately 200 minutes having a maximum leak of about 110 ccm.

(140) A cover **702** was also tested as described above to determine its leak rate. The cover **702** is similar to and includes the components of the cover **406**. For the cover **702**, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having a thickness of about 25 microns. The second adhesive layer **412** comprised a silicone gel, such as a RX1423s silicone gel produced

by Scapa™, having a coating weight of about 150 gsm. The third adhesive layer comprises an acrylic adhesive having a coating weight of about 20 gsm. As shown in FIG. 17, the cover 702 maintained a low leak rate over time. The initial leak rate was slightly greater than 0 ccm, and the leak rate was maintained within a range between about 0 ccm and about 10 ccm for about 200 minutes.

(141) A cover 703 was also tested as described above to determine its leak rate. The cover 703 is similar to and includes components of the cover 406 as described above; however, the cover 703 did not include the third adhesive layer 413. In the example, the film layer 414 was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer 410 was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer 411 was formed from a polyurethane film having thickness of about 25 microns. The second adhesive layer 412 was formed from a polyurethane gel, such as a 9772B polyurethane gel having a coating weight of about 200 gsm. As shown in FIG. 17, the cover 703 maintained a high leak rate between about 0 minutes and about 90 minutes. The leak rate substantially decreased to about 0 ccm as time approached 200 minutes.

(142) Each of the cover 701, the cover 702, and the cover 703, was constructed without a scrim layer. The cover 701, the cover 702, and the cover 703 each exhibited the ability to provide a good seal over a tissue site. Consequently, the construction of the cover 701, the cover 702, and the cover 703, having a film layer, a first adhesive layer, a third adhesive layer, a support layer, and a second adhesive layer, produces a cover suitable for use with negative-pressure therapy while increasing the ability to monitor a tissue site visually without having to remove the cover.

(143) FIG. 18 is a graphical depiction of maceration potential for exemplary covers, illustrating additional details that may be associated with some embodiments. Maceration may refer to the breakdown of skin tissue after prolonged exposure to moisture. In the example, covers were tested to determine each cover's potential to encourage or hinder maceration of healthy tissue. The test apparatus included a substantially smooth surface having electrodes positioned at intervals on the surface. Each cover was placed over the electrodes and sealed to the smooth surface. In the example embodiment, a tissue interface was also placed on the surface, and the cover was sealed to the surface around the tissue interface. Fluid was circulated to the tissue interface to simulate an instillation therapy. In the example embodiment, approximately 180 ml of fluid was instilled into the tissue interface. A current was applied to the electrodes, and the resulting voltage was measured across the electrodes as the fluid is instilled into the tissue interface. If fluid seeps between the cover and the surface surrounding the tissue interface, the resistance between the electrodes can change, indicating the presence of fluid and the potential for maceration. A higher voltage measurement indicates that the moisture content between the surface and the cover is low and, consequently, indicates a lower potential for maceration of the tissue surrounding the tissue site. A lower voltage measurement indicates that the moisture content between the surface and the cover is high and, consequently, indicates a higher potential for maceration of the tissue surrounding the tissue site.

(144) A first cover 801 was tested as described above to determine its maceration potential. The cover 801 is similar to and includes the components of the cover 406 as described above. For the cover 801, the film layer 414 was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer 410 was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer 411 was formed from a polyurethane film having a thickness of about 40 microns. The second adhesive layer 412 was formed from a silicone gel, such as a 6054 silicone gel produced by Scapa™, having a coating weight of about 150 gsm. The third adhesive layer was formed from an acrylic adhesive having a coating weight of about 20 gsm. As shown in FIG. 18, the cover 801 maintained a voltage measurement of slightly less than 5 volts through approximately 50 ml instilled. Across the first 50 ml instilled, the cover 801 exhibited low maceration potential. At approximately 50 ml, the voltage measured began to varying indicating

that as additional fluid was instilled, the potential for maceration increased.

(145) A cover **802** was also tested as described above to determine its maceration potential. The cover **802** is similar to and includes the components of the cover **406** as described above. For the cover **802**, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having a thickness of about 25 microns. The second adhesive layer **412** comprised a silicone gel, such as a RX1423s silicone gel produced by Scapa™, having a coating weight of about 150 gsm. The third adhesive layer comprises an acrylic adhesive having a coating weight of about 20 gsm. As shown in FIG. **18**, the cover **802** maintained a voltage measurement of about 5 volts through approximately 180 ml instilled, indicating that the potential for maceration is minimal.

(146) A cover **803** was also tested as described above to determine its maceration potential. The cover **803** is similar to and includes components of the cover **406** as described above; however, the cover **803** did not include the third adhesive layer **413**. In the example, the film layer **414** was formed from a polyurethane film having a thickness of about 30 microns, and the first adhesive layer **410** was formed from an acrylic adhesive having a coating weight of about 45 gsm. The support layer **411** was formed from a polyurethane film having thickness of about 25 microns. The second adhesive layer **412** was formed from a polyurethane gel, such as a 9772B polyurethane gel having a coating weight of about 200 gsm. As shown in FIG. **18**, the cover **803** the voltage measurement immediately dropped to approximately 0 volts as fluid began to be instilled. The cover **803** maintained the voltage measurement through approximately 180 ml instilled, indicating a high risk for potential maceration.

(147) Each of the cover **801**, the cover **802**, and the cover **803**, was constructed without a scrim layer. The cover **801**, the cover **802**, and the cover **803** each exhibited the ability to provide inhibit maceration of tissue adjacent a tissue site. Consequently, the construction of the cover **801**, the cover **802**, and the cover **803**, having a film layer, a first adhesive layer, a third adhesive layer, a support layer, and a second adhesive layer, produces a cover suitable for use with negative-pressure therapy while increasing the ability to monitor a tissue site visually without having to remove the cover.

(148) The manufacturing processes described herein can produce a cover without the use of a scrim layer. Typically, a scrim layer is needed to provide support for the cover **106**, the cover **206**, the cover **306**, and the cover **406** during the manufacturing process. By manufacturing the cover **106**, the cover **206**, the cover **306**, and the cover **406** without a scrim layer, the light transmittance of the final product can be increased while providing additional support to the cover **106**, the cover **206**, the cover **306**, and the cover **406**, for example, in embodiments having the support layer **111** disposed between the first adhesive layer **110** and the second adhesive layer **112**. Furthermore, the cover **106**, the cover **206**, the cover **306**, and the cover **406** may have improved clarity aiding more accurate cover positioning and visualization of the wound filler and periwound, increased flexibility and conformability, more efficient use of the adhesives, which enable a thinner coating of adhesive, and reduced manufacturing costs. Still further the cover **306** may accommodate shrinkage between the various materials forming the cover **306**.

(149) While shown in a few illustrative embodiments, a person having ordinary skill in the art will recognize that the systems, apparatuses, and methods described herein are susceptible to various changes and modifications. Moreover, descriptions of various alternatives using terms such as “or” do not require mutual exclusivity unless clearly required by the context, and the indefinite articles “a” or “an” do not limit the subject to a single instance unless clearly required by the context. Components may be also be combined or eliminated in various configurations for purposes of sale, manufacture, assembly, or use. For example, in some configurations a dressing, a container, or both may be eliminated or separated from other components for manufacture or sale.

(150) The appended claims set forth novel and inventive aspects of the subject matter described

above, but the claims may also encompass additional subject matter not specifically recited in detail. For example, certain features, elements, or aspects may be omitted from the claims if not necessary to distinguish the novel and inventive features from what is already known to a person having ordinary skill in the art. Features, elements, and aspects described herein may also be combined or replaced by alternative features serving the same, equivalent, or similar purpose without departing from the scope of the invention defined by the appended claims.

Claims

1. A tissue cover including a first film layer having a first side and a second side, a first adhesive layer coupled to the first film layer, and a second adhesive layer coupled to the first adhesive, the second adhesive layer having a plurality of perforations extending through the second adhesive layer, the first film layer, the first adhesive layer, and the second adhesive layer being coextensive, the cover formed by a process comprising the steps of: coupling a first adhesive to the second side of the first film layer to form the first adhesive layer having a first side adjacent to the second side of the first film layer; coupling a second adhesive to a second film layer to form the second adhesive layer having a first side adjacent to the second film layer; forming one or more perforations through the second film layer and the second adhesive layer; coupling a second side of the second adhesive layer to a second side of the first adhesive layer; and removing the second film layer.
2. The tissue cover of claim 1, wherein coupling the first adhesive to the second side of the first film layer comprises casting the first adhesive onto the first film layer.
3. The tissue cover of claim 1, wherein coupling the first adhesive to the second side of the first film layer comprises laminating the first film layer to the first adhesive.
4. The tissue cover of claim 1, wherein coupling the first adhesive to the second side of the first film layer further comprises curing the first adhesive.
5. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises punching one or more perforations through the second film layer and the second adhesive layer.
6. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises cutting one or more perforations through the second film layer and the second adhesive layer.
7. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises melting one or more perforations through the second film layer and the second adhesive layer.
8. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises drilling one or more perforations through the second film layer and the second adhesive layer with a rotary tool.
9. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises forming each perforation with an average effective diameter between about 2 mm and about 50 mm.
10. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises forming each perforation with an average effective diameter of about 30 mm.
11. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises forming each perforation with an average effective diameter of about 7 mm.
12. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises forming one or more perforations so that adjacent perforations have centers separated by between about 5 mm and about 100 mm.

13. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises forming one or more perforations so that adjacent perforations have centers separated by about 10 mm.
 14. The tissue cover of claim 1, wherein forming one or more perforations through the second film layer and the second adhesive layer comprises removing about 50% of a surface area of the second film layer and the second adhesive layer.
 15. The tissue cover of claim 1, wherein the second adhesive comprises a silicone gel.
 16. The tissue cover of claim 15, wherein the silicone gel is a two-part silicone system and the process further comprises: providing a first part of the two-part silicone system; providing a second part of the two-part silicone system; mixing the first part with the second part to form a two-part silicone gel; and curing the two-part silicone gel.
 17. The tissue cover of claim 15, wherein the silicone gel is a one-part silicone.
 18. The tissue cover of claim 1, wherein coupling the first side of the second film layer to the second side of the first adhesive layer comprises laminating the first side of the second film layer to the second side of the first adhesive layer.
 19. The tissue cover of claim 1, wherein coupling the second adhesive to the second film layer is performed without use of a scrim layer.
 20. The tissue cover of claim 1, wherein coupling the first adhesive to the first film layer is performed without use of a scrim layer.
 21. The tissue cover of claim 1, wherein: coupling the first adhesive to the first film layer is performed without use of a scrim layer; and coupling the second adhesive to the second film layer is performed without use of a scrim layer.
 22. The tissue cover of claim 1, wherein the second adhesive layer has a coating weight between about 150 gsm and 250 gsm.
 23. The tissue cover of claim 1, wherein the first adhesive layer has a coating weight between about 10 gsm and about 60 gsm.
 24. The tissue cover of claim 1, wherein the second adhesive further comprises a platinum catalyst.
 25. The tissue cover of claim 1, wherein the second adhesive further comprises a sulfur catalyst.
 26. The tissue cover of claim 1, further comprising forming a sizing line through the combined first film layer, the first adhesive layer, and the second film layer.
 27. The tissue cover of claim 26, wherein the sizing line comprises a plurality of sizing lines equidistantly spaced from each other across the tissue cover.
 28. The tissue cover of claim 26, wherein forming the sizing line comprises forming a plurality of perforations through the first film layer, the first adhesive layer, and the second film layer.
 29. The tissue cover of claim 28, wherein the perforations comprise slits spaced about 25 mm to about 60 mm from adjacent slits.
 30. The tissue cover of claim 28, wherein the plurality of sizing lines are between about 5 cm and about 30 cm from adjacent sizing lines.
 31. The tissue cover of claim 28, wherein the plurality of sizing lines are about 18 cm from adjacent sizing lines.
 32. The tissue cover of claim 1, wherein a transmittance of the cover is between about 70% and about 100%.
 33. The tissue cover of claim 1, wherein a transmittance of the cover is about 90%.
 34. The tissue cover of claim 1, wherein a color perceived through the cover is substantially the same as the color perceived independently of the cover.
 35. The tissue cover of claim 1, wherein the process further comprises the step of removing the second film layer.
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